

Zscaler Digital Delivery Consultant Practice Lab Guide



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Zscaler Digital Delivery Consultant Practice Lab Guide

January, 2025, Rev. 1.1

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About the Delivery Consultant Practice Labs

Welcome to the delivery Consultant Practice Lab. This lab provides an opportunity to practice & refresh your ZIA skills based on common deployment scenarios. The hands-on lab will be facilitated by an instructor who is qualified and experienced in deploying ZIA,ZPA,ZDX . After the practice session, you will be assigned a typical deployment scenario and an unconfigured lab environment. You will be given the time needed to implement a solution that meets the requirements set out in the scenario. Your solution will be evaluated against a checklist of requirements and a reference implementation. A mark of pass or fail will be awarded. A pass is required to be certified as a Delivery Consultant. Specialist.

Scenario

Consider the scenario where Safemarch, a large global organization is transforming their security and network operations. To transition to a Zero-Trust Network architecture, Safemarch has decided to deploy Zscaler Internet Access (ZIA), Zscaler Private Access (ZPA) and Zscaler Digital Experience (ZDX) through Zidentity.

You now need to apply what you have learned to do further preparation for the rollout of Zscaler services across Safemarch.

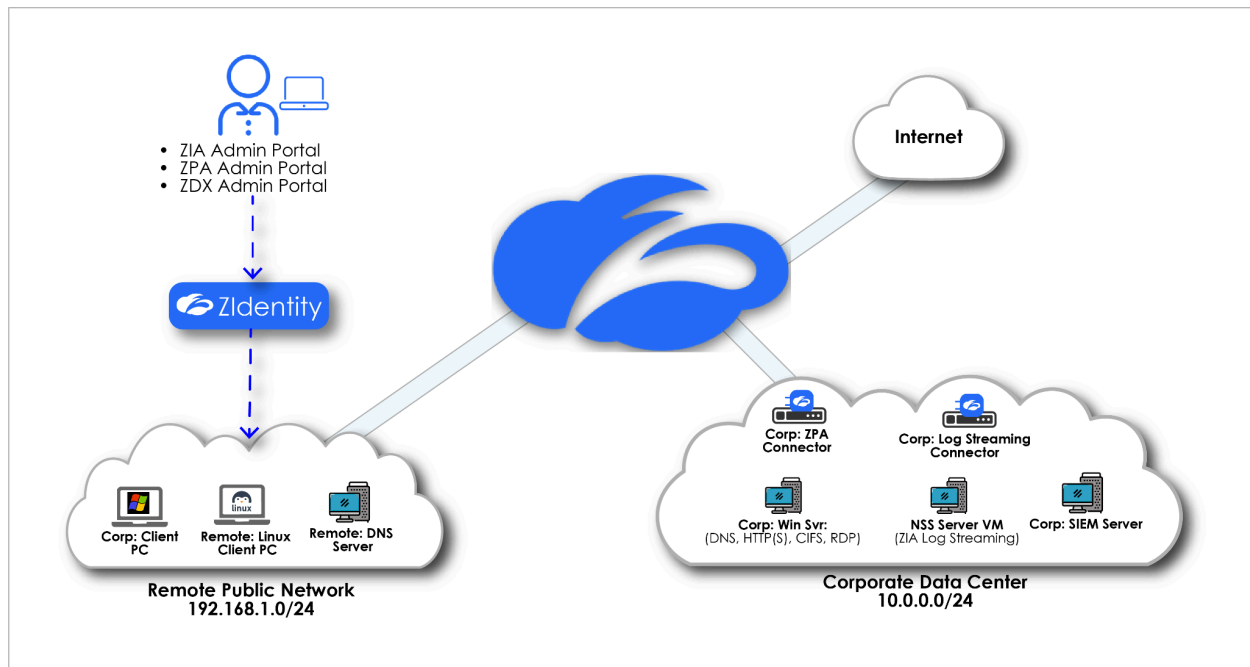
Requirements for items that need to be configured include:

- Zidentity admin entitlements for administrators to fully deploy and support users.
- Zidentity service entitlements for end users to access needed services.
- Traffic Forwarding via Zscaler Client Connector and Zscaler App Connectors.
- Log Streaming & Reporting.
- Digital Experience Monitoring for users.
- Policies that enforce secure, policy-based access of employees to Safemarch internal applications.
- Privacy for Safemarch users accessing their private information while ensuring that all other traffic is inspected for threats and data leakage.

Lab Setup

For this lab each student is assigned a unique student user (e.g. student@zSabcd.safemarch.com)for access to a dedicated training pod of virtual machines (hosted in Skytap) with corresponding tenants.

Virtual Lab network topology



Name	IP	Username	Password	Description
Corp: Client PC	192.168.1.1	Administrator	Admin-123!	Windows client
Remote: Linux Client PC	192.168.1.2	Admin	Admin-123!	Linux client
Remote: DNS Server	192.168.1.252	Administrator	Zscaler123!	DNS Server
Corp: Win Svr	10.0.0.8	Administrator	Admin-123!	Windows Server
Log streaming connector	10.0.0.20	admin	zscaler	App connector as ZPA log streaming
NSS Server	10.0.0.4	zsroot	zsroot	ZIA log streaming
Corp: SIEM Server	10.0.0.2	zscaler	Zscaler123!	Syslog server
ZPA Connector	10.0.0.3	admin	zscaler	Private app connectivity

Lab 1: Configure Identity Services

In this task, you need to create organization specific Departments and Users with email addresses to be used for addressing administrative and user related communication

Task Configuration Requirements

1. Create departments, add users to respective departments.
2. Configure Zscaler Support Remote Assistance to view-only access for 90 days to allow Zscaler support to observe the portal's status and data during troubleshooting

Use the below table to create the Departments and the Users:

User Name	Password	Department
marketing- <code><cd></code> @ zsabcd .safemarch.com	Zscaler123!	Marketing
finance- <code><cd></code> @ zsabcd .safemarch.com	Zscaler123!	Finance
itsecurity- <code><cd></code> @ zsabcd .safemarch.com	Zscaler123!	IT Security
admin- <code><cd></code> @ zsabcd .safemarch.com	Zscaler123!	Administrator
maintenance- <code><cd></code> @ zsabcd .safemarc <code><cd></code> ch.com	Zscaler123!	Maintenance

Note: 1. Use the **Student Admin** for the **Assign Groups** and enable the **Skip Second Factor Authentication** option in the **Security Settings** to bypass MFA.

2. Ensure the Primary email address is set to be the user (**student@zsabcd.safemarch.com**) assigned to you in the Lab Access Information

3. Make sure to select the "**Set by Administrator**" option in the password option under **Security Settings**

4. Make sure to uncheck the option "**Prompt for Password Change After the Initial Log In**".

Where to get Help

- <https://help.zscaler.com/zidentity/configuring-authentication-session>
- <https://help.zscaler.com/zidentity/enabling-remote-assistance>

Lab 2: Configure ZIA and ZPA Traffic Forwarding

In this lab, you need to prepare traffic forwarding configurations for Safemarch's Zscaler Client Connector users.

Configuration Requirements

set up traffic forwarding to meet Safemarch's requirements, including:

1. Ensure when users are working from a remote location, all traffic from all ports and protocols, must be able to connect to Zero trust exchange.
2. Users must use a password to disable ZIA and ZPA.
3. Client Connector Log Mode must be "Warn".
4. Client Connector must automatically install the Zscaler SSL certificate.
5. Client Connector configuration needs to be set to potentially allow traffic on any port and protocol to be forwarded.
6. Change ZPA Reauthentication time to 10 minutes.

Testing Requirements

- Login to Client PC VM, use student@zsabcd.safemarch.com account to login to Zscaler Client Connector and verify in Web Insights logs that the user is able to access the Internet.

Where to get Help

If you need help with the details of each of the Zscaler Client Connector procedures, you can use the following links:

- [Configuring Forwarding Profiles for Zscaler Client Connector](#)
- [Configuring Zscaler Client Connector Profiles](#)

Lab 3: Configure ZIA Policy

In this lab, you need to configure a set of policies that provide protection against security threats and control access to web-based applications, based on department membership.

Note: Rule order is important, rules which are higher in order will be inspected first.

Configuration Requirements

Following recommended policy settings, implement a set of policies for the initial Safemarch deployment, including:

1. SSL inspect all traffic
2. Exempt some specific web traffic from SSL inspection, including:
 - a. Health URL Categories to address privacy concerns
 - b. A custom list of URLs that may be maintained by adding / removing URLs as needed to respond to SSL inspection-related issues.
 - c. Domains used by native applications that use certificate pinning: (Refer to <https://help.zscaler.com/zia/certificate-pinning-and-ssl-inspection>)
 - Adobe Acrobat Reader (Fill & Sign)
 - Apple iMessage
3. Caution users in the IT Security department for Executable file downloads from any URL category. Also Block Executable file download for rest of users.
4. Block the general file type of "Undetectable" for all users.
5. Finance users should be blocked for uploading all files except Microsoft office files.
6. All traffic must be firewall inspected. Add specific rules to allow access for:
 - a. An approved list of network services which includes SSH for users in the IT Security department.

7. All users need to be granted access to all URL Categories, except for a default rule to block Legal Liability URL Categories that includes:
 - Adult Material, Drugs, Gambling, Illegal or Questionable, Militancy/Hate and Extremism
 - a custom URL category for any additional URLs that may be deemed off limits by Safemarch security and access policies. (Hint: Add any arbitrary URL's as part of this new custom URL Category)
 - b. The customer's IT Security team needs to be able to get access to www.eicar.org site file downloads for continuous evaluation of security tools' effectiveness. Also Block www.eicar.org for the rest of the users.
 - c. Block all users from using Social Networking, EXCEPT employees in the Marketing department must be allowed to view and post to LinkedIn
 - d. Office 365: Forward all O365 traffic and enable Microsoft Recommended One-Click configuration.
 - e. Enable advanced policy settings for URL & Cloud App Controls, including:
 - i. Suspicious New Domains Lookup
 - ii. AI/ML based Content Categorization
 - iii. Embedded Sites Categorization
 - iv. Enforce SafeSearch
8. Allow login to only the Microsoft O365 Safemarch corporate tenant and prevent personal login to Microsoft O365 products like OneDrive and Outlook.
 - a. Make sure you add an SSL inspection rule for MS Login Services to allow the Tenant Profile. Ensure that this particular SSL Inspection rule is higher in order than the "Office 365 One Click" rule.
 - b. Tenant Profile Name: **Safemarch Microsoft O365**

Tenant Directory: **deadbeef-0001-fee-0001-314159265358**

Office 365 Tenant: **safemarch.com**

Testing Requirements

- Check Web Insights logs and certificate info on various URLs to confirm SSL inspection/exemptions
- Verify access to some legal liability URL, e.g. www.freelots-machines.com is blocked
- Verify access to social networking sites is blocked for users, not in the Marketing department
- Check Firewall logs and make sure SSH traffic is hitting the intended firewall rule
- Attempt to go to office.com and search weblogs to make sure there are no entries

Where to get Help

If you need help with the details of each of the policy configuration procedures, you can use the following links:

- [Configuring SSL Inspection Policy](#)
- [Configuring Custom URL Categories](#)
- [URL Format Guidelines](#)
- [Configuring Firewall Policies](#)
- [Configuring URL Filtering Policy](#)
- [Adding Rules to the Cloud App Control Policy](#)

Configuration Hints

1. When configuring a custom list of URLs to respond to SSL inspection-related issues, add some URL, e.g. example.com.

Lab 4: Configure ZIA Logging & Reporting

In this lab, you need to configure Nanolog Streaming Service (NSS) to accommodate Safemarch's need for long-term storage of and real-time streaming of web logs to a SIEM.

Note: In this scenario, a NSS Server VM (10.0.0.4) and a SIEM Server VM (10.0.0.2) have already been pre-installed in your Skytap environment.

Configuration Requirements

configure log streaming to meet Safemarch's logging requirements to send logging information to an onsite SIEM for long-term retention and integration into the overall security information infrastructure. For this deployment you will initially configure ZIA to stream logs to a SIEM server to test the configuration. You will configure to stream the following log data:

- a. Web logs (CSV format)
- b. Include only the records for web transactions that have a Block policy applied for any reason.

Testing Requirements

- On the Corp Client VM, generate traffic that matches the NSS configuration.
- On the SIEM Server VM, verify that log messages are being received in `/var/log/syslog`

Where to get Help

If you need help with the details of each of the required NSS configuration procedures, you can use the following link:

- [Adding NSS Feeds](#)
- [NSS Deployment Guide for VMware vSphere](#)

Configuration Hints

- Start the syslog, check the status.
 - `sudo systemctl start rsyslog`
 - `sudo systemctl status rsyslog`
- <https://www.linkedin.com/pulse/how-install-set-up-rsyslog-server-linux-ubuntu-20041-akshay-sharma>
- View the syslog entries as they are received.
- When configuring the NSS Server's **Management Interface** IP address, use: **10.0.0.4/24**
- When configuring the NSS Server's **Service Interface** IP address, use: **10.0.0.5/24**
- For the **Nameserver** IP address, use **10.0.0.254**
- For the **Default Gateway** IP address, use **10.0.0.254**
- TCP Port use **514**
- To transfer the NSS Certificate to the NSS Server VM, you can use **WinSCP** which is pre-installed on the WinSvr.

Lab 5: Configure ZPA Policy

Note: Rules higher in order will be implemented first. Once a rule gets 'hit', next rules are not considered.

Configuration Requirements

Following recommended policy settings, implement a set of policies for the initial Safemarch deployment, including:

1. Provision App connector.
2. Allow all employees access to the following private applications:
 - a. Intranet: intranet.pattraining.safemarch.com, Use TCP ports 80 and 443
 - b. File Share: shares.pattraining.safemarch.com, Use TCP ports 139 and 445
3. Allow access to a SIEM server via **SSH** only from devices on domain-joined devices. User TCP ports 22. (Hint: Use Device Posture in Client Connector Portal.)
4. Only users in the **Maintenance** department should be able to access **hvac.pattraining.safemarch.com**. Use TCP ports 80.
 - a. Ensure that idle, open connections are closed after 15 min.
5. All users EXCEPT users from the Administrator Department are to be blocked from using Safemarch's remote desktop application: **rdp.pattraining.safemarch.com**. Use TCP and UDP port 3389.
 - a. RDP connections require re-authentication every 10 min.

Username: Administrator

Password: Admin-123!
6. Safemarch has identified a list of applications that should never be accessed through ZPA, including: (Use Client Forwarding Policy)
 - a. erp.safemarch.com
 - b. crm.safemarch.com
 - c. contractors.safemarch.com
 - d. b2b.safemarch.com

7. Auto-discover any other applications that are running on the Safemarch corporate network.

Testing Requirements

- Check the status of the App Connector.
- To test access to the corporate file share application, access `\\shares.pattraining.safemarch.com` from the Client PC
- To test RDP access, open an RDP session from the Windows taskbar by typing **rdp** and connect to rdp.pattraining.safemarch.com
- Check **Diagnostics** logs and **Live** Logs in the ZPA Admin Portal to verify that your policies have the desired effect.

Where to get Help

If you need help with the details of each of the policy configuration procedures, you can use the following links:

- [Configuring Application Segments](#)
- [Configuring Segment Groups](#)
- [Configuring Server Groups](#)
- [Configuring Access Policies](#)
- [Configuring Timeout Policies](#)

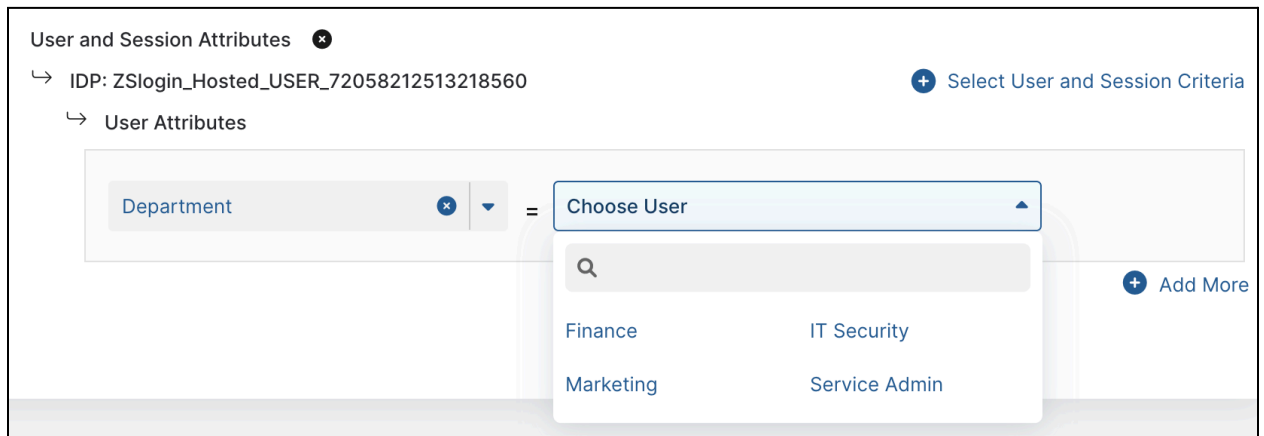
Configuration Hints

- To transfer the provisioning key to the App Connector, you can use WinSCP on the Corp:Win Svr VM.
- For the purposes of this lab only, it is sufficient to enable one App Connector in the data center.

Note: If the provisioning of an App Connector fails, re-provision it as follows:

1. Stop the App Connector service using the command **sudo systemctl stop zpa-connector**
2. Delete all files in the relevant folder with the command **sudo find "/opt/zscaler/var" -mindepth 1 -delete**
3. Reboot the VM using the command **reboot**
4. Stop the App Connector service again (it auto-starts on boot) using the command **sudo systemctl stop zpa-connector**
5. Copy back the Provisioning Key file using the command **sudo cp provision_key /opt/zscaler/var/**
6. Start the App Connector service again using the command **sudo systemctl start zpa-connector**
7. Check App Connector status using the command **sudo systemctl status zpa-connector**

- To select the Departments while adding policies, Click **Add Criteria**. Select User and Session Attributes. Only then you can add Departments as shown below.



Lab 6: Configure ZPA Logging & Reporting

In this lab, you need to configure Log Streaming Service (LSS) to accommodate Safemarch's need for long-term storage of and real-time streaming of web logs to a SIEM.

- A SIEM Server VM has already been pre-installed in your environment.

Configuration Requirements

configure log streaming to meet Safemarch's logging requirements to send logging information to an onsite SIEM for long-term retention and integration into the overall security information infrastructure.

1. Configure the App Connector **Corp: Log Streaming Connector**. This is used for streaming logs from ZPA cloud to SIEM server.
2. You will configure LSS to stream the following log data:
 - a. User Activity logs for Client Connector users only (CSV format)
 - b. App Connector Status logs (CSV format)

Testing Requirements

- On the Corp: Client PC, generate traffic that matches the LSS configuration.
- Log in to the SIEM server VM and verify that log messages are being received.
- Ensure SIEM server can be accessed via ZPA.

Where to get Help

If you need help with the details of each of the required LSS configuration procedures, you can use the following link:

- [Configuring a Log Receiver](#)

Configuration Hints

- To upload the provisioning key to the App Connector, you can use **WinSCP** which is pre-installed on the **Corp: Win Svr**

Note: If the provisioning of an App Connector fails, re-provision it as follows:

1. Stop the App Connector service using the command **sudo systemctl stop zpa-connector**
2. Delete all files in the relevant folder with the command **sudo find "/opt/zscaler/var" -mindepth 1 -delete**
3. Reboot the VM using the command **reboot**

4. Stop the App Connector service again (it auto-starts on boot) using the command **sudo systemctl stop zpa-connector**
5. Copy back the Provisioning Key file using the command **sudo cp provision_key /opt/zscaler/var/**
6. Start the App Connector service again using the command **sudo systemctl start zpa-connector**
7. Check App Connector status using the command **sudo systemctl status zpa-connector**

- The TCP Port for the Log Receiver is **514**.

Lab 7: Protect the leaking of HIPAA documents

In this lab, you will learn how to protect sensitive HIPAA-related documents from being leaked during file upload activities by configuring Data Loss Prevention (DLP) policies in Zscaler Internet Access (ZIA). The goal is to ensure that all file upload traffic is inspected, medical information containing HIPAA-sensitive data is blocked, and activities are logged.

Configuration Requirements

implement a set of configurations to ensure the protection of sensitive HIPAA data during file uploads. This includes:

1. Ensure that all file upload traffic is fully inspected to ensure that Medical Information with HIPAA sensitive data is blocked and the activity logged.

Testing Requirements

1. Login with the student@zsabcd.safemarch.com credentials to Zscaler Client Connector on the Corp: Client PC and verify that Internet Security status is enabled in the Zscaler Client Connector.
2. Open <https://www.dochub.com/en/main-features/upload-a-pdf-document> in the browser and confirm access and traffic is being inspected.
3. Attempt to upload the Medical Info pdf from the DLP Test Files folder on the desktop and verify that the uploading is blocked
4. To make sure the HIPAA file **Medical Info** is getting blocked by the policy.

Where to Get Help

Refer to the following resources for guidance on configuring and testing DLP policies:

- [Configuring DLP Policies](#)
- [Managing DLP Dictionaries and Engines](#)
- [SSL Inspection Configuration](#)
- [Testing DLP Policy Effectiveness](#)

Configuration Hints

1. The built-in HIPAA DLP dictionary may be used.
2. It may be necessary to adjust the Confidence Score Threshold on the included DLP dictionaries if uploading is not blocked on the test documents provided

Lab 8: Adding Custom Application and Creating Custom Probes

The Applications page in the ZDX admin portal provides an overview of the configured applications. The Zscaler service provides predefined applications or you can configure a customized application. Each application is designed to have corresponding probes associated with it.

Configuration Requirements

Add a custom application in the ZDX Admin portal as per the below details:

Name of the Custom Application	Type	Status
Test Application	Custom	Enabled

Add probe to the Custom Applications as per the details as mentioned below:

Web Probe	
General	
Type of the Probe	Web Probe
Name of the Probe	Test_Web
Status	Enabled
Application	Test Application
Run Frequency	5 Mins

Additional Parameters	
HTTP Response Status Codes	Successful Response (200-299), 304(Not Modified)
Number of Attempts	2
Timeout (seconds)	60
Follow Redirect	Enable
Maximum Redirects	5

Cloud Path Probe	
General	
Type of the Probe	Cloud Path Probe
Name of the Probe	Test_CloudPath
Status	Enabled
Follow Web Probe	Test_Web
Application	Test Application
Run Frequency	5 Mins
Cloud Path Probe Configuration	
Protocol	ICMP
Packet Count	5
Interval	1000ms
Timeout	1000ms

Note: You can choose Destination URL(for Web Probe) and the Cloud Path Host (for the Cloud Path Probe) basis your choice

Where to get Help

- <https://help.zscaler.com/zdx/adding-custom-application>
- <https://help.zscaler.com/zdx/configuring-probe>
- <https://help.zscaler.com/zdx/about-probes>

Lab 9: Creating Custom Alert

Alerts provide a primary and timely source of information to monitor device, application, network performance, and ZDX Score on the Alerts page so that you can analyze and remediate issues. In the ZDX Admin Portal, you can view alerts triggered over the past two weeks. You can select options from 2 hours to 14 days in the time range filter to view triggered alerts in the Alert History tab.

Configurational Requirements

Create custom alerts in ZDX Admin portal the based on the details below:

Alert Name	Alert Type (App/Network / Device/ZDX Score)	Severity	Filters	Criteria	Action
TestApp_Alert	Application	High	Application: Test Application Probe: TestApp Probe Filters: Geolocation: Europe	Page Fetch Time >= 80 0 ms ZDX Score < 10	Alert Only if Repeated: 3 times in a Row Delivery Method: Email

Note: For the Alerts to be created successfully the user needs minimum 1 active device against the application.

For the **Delivery Method** as **Email** you can input the student credentials (student@zsabcd.safemarch.com) assigned to you.

Where to get Help

- <https://help.zscaler.com/zdx/about-alerts>
- <https://help.zscaler.com/zdx/triggering-alert>

Task 10: Apply Zero Trust Automation Requirements: Adding URL Filtering Policy

Configure a OneAPI client that is able to be used in an automation that creates a URL filtering rule that blocks the gambling websites.

Configuration Requirements

Configure a OneAPI client that is able to be used in an automation that creates a URL filtering rule that blocks the gambling websites.

Testing Requirements

1. Create an API client in ZIdentity Dashboard (Use the Corp: Win Svr VM)

Note: a. Use the Corp: Win Svr VM to download the Postman client (<https://www.postman.com/downloads/>)

b. Install the Postman Client in the Corp: Win Svr and signup for the new user

2. Authenticate a session for the OneAPI client in Postman
3. Call the API endpoint to create the URL filtering policy
4. Navigate to the ZIA Admin portal and see if the URL filtering rule is created
5. Open a browser and navigate to any gambling website (e.g. <https://www.888.com>) to check if the navigation is blocked by the URL filtering policy.

Note: While creating the JSON payload to call the API endpoint for adding a URL Filtering Rule ensure passing the parameter for Protocol Criteria as "protocols": ["ANY_RULE"]

Where to get Help

If you need help with the details of each of the policy configuration procedures, you can use the following links:

- <https://help.zscaler.com/oneapi/understanding-oneapi-endpoints>
- <https://help.zscaler.com/oneapi/configuring-postman-rest-api-client>
- <https://help.zscaler.com/zia/3rd-party-app-governance-api>
- <https://help.zscaler.com/zia/url-filtering-policy#/urlFilteringRules-post>

Note: For the step-by-step detailed approach to access the API endpoints refer to the **API Configuration Guide** attached with this Lab environment.